

AISC ANSWER KEY

MODULE 1:

1. Write short notes on Applications of AI.

1. AI in Marketing

AI analyzes customer behavior and preferences to recommend products and services. It uses predictive analytics to provide personalized suggestions. Example: Netflix recommends movies and shows based on viewing history.

2. AI in Banking

AI-powered chatbots provide 24×7 customer support and answer customer queries instantly. They improve customer experience by reducing response time. Example: HDFC Bank's EVA chatbot handles millions of customer interactions.

3. AI in Gaming

AI creates intelligent game opponents that can learn and make strategic decisions. It improves gameplay by adapting to player actions. Example: AlphaGo defeated world champion Lee Sedol in the game of Go.

4. AI in Space Exploration

AI is used for processing satellite images, monitoring spacecraft systems, and assisting astronauts. It also supports satellite and spacecraft development. AI helps improve the efficiency and accuracy of space missions.

2. Give PEAS descriptors for Vacuum cleaner.

Performance Measure: This defines how the success of the agent is judged. It includes cleanliness of the floor, efficiency of cleaning, minimal energy/battery consumption, and reduced noise.

Environment: This is the setting in which the agent operates and perceives. It includes the room, floor type (carpet/tile), furniture, walls, and dirt distribution.

Actuators: These are the components through which the agent acts on the environment. They include wheels for movement, brushes for scrubbing, and a suction motor to collect dirt.

Sensors: These are the components through which the agent perceives its environment. They include dirt detection sensors, bump/obstacle sensors, and cliff sensors (to avoid falling off stairs).

3. PEAS for Self-Driving Car Agent

Performance Measure: This defines how the success of the agent is judged. It includes safety (avoiding accidents), speed/time to reach destination, fuel/energy efficiency, passenger comfort, and obeying traffic rules.

Environment: This is the setting in which the agent operates and perceives. It includes roads, traffic signals, other vehicles, pedestrians, lane markings, and weather conditions.

Actuators: These are the components through which the agent acts on the environment. They include steering wheel, accelerator, brakes, gear system, and signal/indicator lights.

Sensors: These are the components through which the agent perceives its environment. They include cameras, GPS, speedometer, LIDAR/radar, and proximity/ultrasonic sensors.

4. Explain WUMPUS world environment giving its PEAS description / Note on Wumpus World Environment.

Wumpus World is a 4×4 grid-based environment where an agent must find gold while avoiding the Wumpus and deadly pits, using only percepts.

PEAS Description:

- **Performance Measure:** This defines how the success of the agent is judged.
+1000 for gold, −1000 for death, −1 per action, −10 for shooting an arrow.
- **Environment:** This is the setting in which the agent operates and perceives.
4×4 grid containing the agent, Wumpus, gold, and pits.
- **Actuators:** These are the components through which the agent acts on the environment.
Move forward, turn left/right, grab gold, and shoot arrow.
- **Sensors:** These are the components through which the agent perceives its environment.
They include percepts like Stench (near Wumpus), Breeze (near pit), Glitter (on gold), Bump (hit wall), and Scream (Wumpus killed).

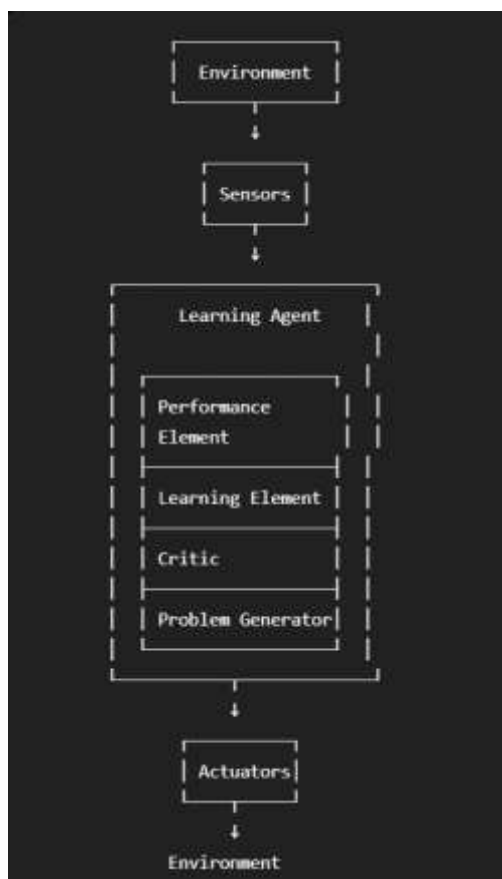
5. Describe different types / properties of environment of AI agents.

Properties of Task Environment

1. **Fully Observable / Partially Observable:** Fully observable if the agent can access the complete state; otherwise, it is partially observable.
2. **Deterministic / Stochastic:** Deterministic if the next state is completely determined by the current state and action; otherwise, it is stochastic.
3. **Episodic / Sequential:** Episodic if each action is independent; sequential if current actions affect future decisions.
4. **Static / Dynamic:** Static if the environment does not change while the agent is deciding; dynamic if it can change during that time.

6. Explain the Learning Agent with suitable block diagram.

A learning agent is an AI agent that learns from its experiences and feedback and improves its performance over time. It continuously updates its knowledge based on feedback to make better future decisions.



MODULE 2:

1. What do you mean by state space representation? Explain with example of it.

1. State Space Representation

State space representation is a method of representing an AI problem using a set of all possible states of the problem and the actions/operators that can transform one state into another.

It consists of:

1. Initial State – The starting condition of the problem.
2. Goal State – The desired final condition.
3. Operators/Actions – Actions that change one state into another.
4. State Space – The collection of all possible states reachable from the initial state.

Example: 8-Puzzle Problem

In the 8-puzzle, the board contains 8 numbered tiles and one blank space.

Initial State	Action	Goal State
1 2 3 4 _ 6 7 5 8	Move tile 5 into blank	1 2 3 4 5 6 7 8 _

- Initial state: 1 2 3 / 4 _ 6 / 7 5 8
- Goal state: 1 2 3 / 4 5 6 / 7 8 _
- Operators: Move the blank tile up, down, left, or right.
- State space: All possible arrangements of the tiles that can be reached from the initial state.

2. Enumerate Classical 'Water jug Problem'. Describe the state space for this problem and also give the solution.

8. Classical Water Jug Problem

The **Water Jug Problem** is a classical AI problem in which we have two jugs of fixed capacities and must measure a specific quantity of water using operations like **fill**, **empty**, and **pour**.

Example:

Consider two jugs of **4 litres** and **3 litres**. The goal is to obtain **2 litres of water** in the 4-litre jug.

State Space Representation:

A state is represented as (x, y) , where:

- x = amount of water in the 4-litre jug
- y = amount of water in the 3-litre jug

Initial state: $(0, 0)$

Goal state: $(2, 0)$ or $(2, y)$

Possible operations:

1. Fill either jug completely.
2. Empty either jug completely.
3. Pour water from one jug into the other until the first is empty or the second is full.

Solution:

Step	State (4L, 3L)	Operation
1	$(0, 0)$	Initial state
2	$(4, 0)$	Fill 4L jug
3	$(1, 3)$	Pour 4L $\rightarrow$ 3L
4	$(1, 0)$	Empty 3L jug
5	$(0, 1)$	Pour 4L $\rightarrow$ 3L
6	$(4, 1)$	Fill 4L jug
7	$(2, 3)$	Pour 4L $\rightarrow$ 3L
8	$(2, 0)$	Empty 3L jug — Goal reached

Therefore, 2 litres of water are obtained in the 4-litre jug.

3. What do you understand by uninformed search methods? Explain BFS (Breadth First Search) and DFS (Depth First Search) with suitable example.

9. Uninformed Search Methods

Uninformed search methods are search techniques that have no additional information about the goal. They use only the initial state, operators, and goal test to explore the state space.

1. Breadth First Search (BFS)

- BFS explores the nodes level by level, starting from the root node.
- It uses a FIFO (First-In-First-Out) queue.
- BFS is complete and gives the shortest path when all step costs are equal.

Example:



BFS order: A → B → C → D → E → F

2. Depth First Search (DFS)

- DFS explores a path as deeply as possible before backtracking.
- It uses a LIFO stack or recursion.
- DFS requires less memory but may not find the shortest path.

Example:

For the same tree:

DFS order: A → B → D → E → C → F

Difference:

BFS	DFS
Explores level by level	Explores depth-wise
Uses Queue	Uses Stack
More memory	Less memory
Finds shortest path (equal costs)	Does not guarantee shortest path

4. What is formulation of a problem. Formulate the Wumpus world problem in terms of initial state, actions, successor function, goal test, path cost.

Problem formulation is the process of defining a problem in terms of its initial state, actions, successor function, goal test, and path cost so that an AI agent can find a solution.

Wumpus World Problem Formulation

1. Initial State:

The agent starts at square (1,1), with the Wumpus, pits, and gold placed at unknown locations.

2. Actions:

The agent can Move Forward, Turn Left, Turn Right, Grab, Shoot, and Climb.

3. Successor Function:

It describes the state resulting from each action. For example, moving forward changes the agent's position, while Grab picks up the gold.

4. Goal Test:

The goal is achieved when the agent has the gold and safely climbs out of the cave.

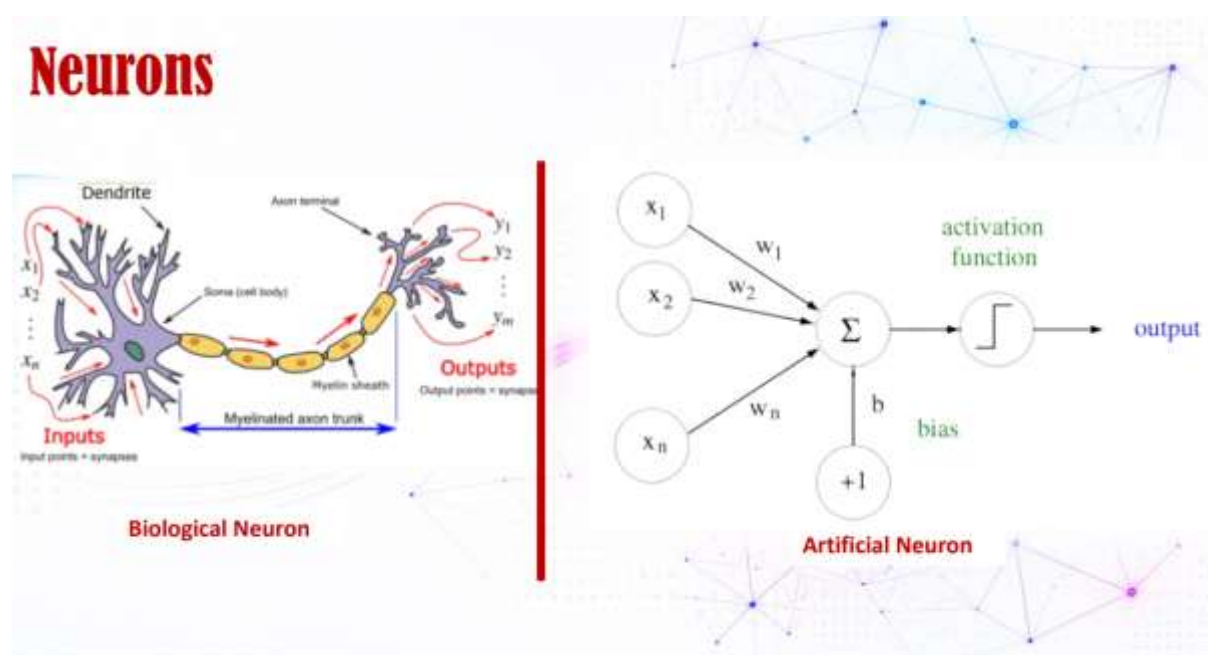
5. Path Cost:

Each action has a cost, and the objective is to reach the goal with minimum total cost while avoiding pits and the Wumpus.

MODULE 5

1. Differentiate between biological and artificial neurons with the help of labelled diagrams.

Biological Neuron	Artificial Neuron
1. It is a natural information-processing unit found in the human brain.	1. It is a computational model inspired by biological neurons.
2. It consists of dendrites, cell body and axon .	2. It consists of inputs, weights, bias, activation function and output .
3. Dendrites receive signals from other neurons.	3. Inputs receive data from the external environment or previous neurons.
4. It communicates using electrochemical signals .	4. It processes information using mathematical calculations .
5. Connections between neurons are formed through biological synapses.	5. Connections between neurons are represented using weights .



2. Explain the structure and working of a biological neuron with the help of a labelled diagram.

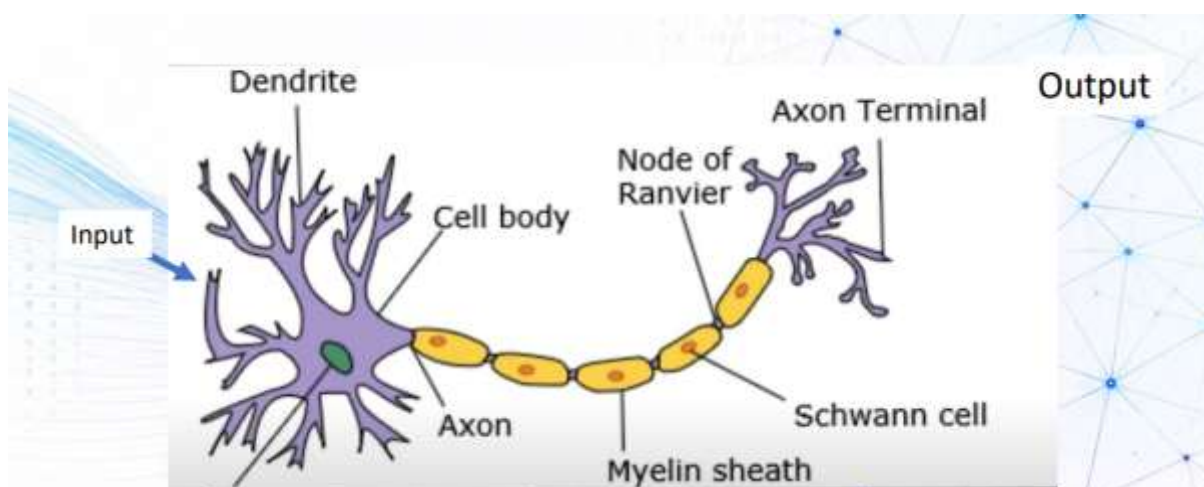
Structure of a Biological Neuron

1. **Dendrites:**
Dendrites receive information/signals from other neurons or from the surrounding environment.
2. **Cell Body (Soma):**
It contains the nucleus and performs the main processing and maintenance functions of the neuron.
3. **Axon:**
The axon carries the processed electrical signal away from the cell body toward the end of the neuron.
4. **Myelin Sheath:**
It surrounds the axon and helps in efficient transmission of signals along the axon.
5. **Axon Terminals:**
These are the endings of the axon that transmit the signal to other neurons.

Working of a Biological Neuron

1. **Receiving signals:** Dendrites receive information from other neurons.
2. **Processing:** The received information is passed to the cell body, where it is processed.
3. **Signal transmission:** If the neuron generates a signal, it travels through the axon.
4. **Passing information:** The signal reaches the axon terminals and is transmitted to the next neuron.
5. **Information flow:** Thus, the general flow of information is:

Dendrites → Cell Body → Axon → Axon Terminals → Next Neuron



3. Illustrate the architecture of an artificial neuron and explain the function of each component.

Components and their Functions

1. Inputs (x_1, x_2, x_3):

Inputs are the information/data supplied to the artificial neuron. A neuron can receive multiple inputs.

2. Weights (w_1, w_2, w_3):

Each input is multiplied by an associated weight. Weights determine the importance or strength of each input. During training, these weights are updated. A higher weight indicates that the input is considered more important.

3. Weighted Sum (Σ):

The neuron calculates the sum of all weighted inputs:

4. Bias (b):

Bias is a constant added to the weighted sum. It helps shift the result to the positive or negative side and changes the range of the weighted input.

5. Activation Function:

After the linear combination of inputs, weights and bias is obtained, an activation function is applied. It converts the processed input into the final output and introduces non-linearity.

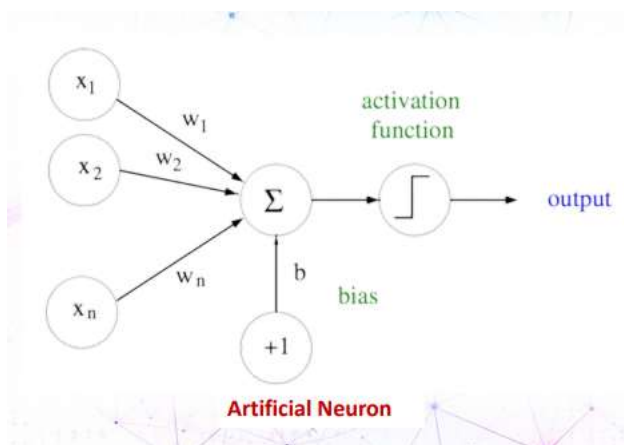
6. Output (y):

The output is the final result produced by the neuron after applying the activation function. It may be passed to another neuron or may represent the final output.

Working of Artificial Neuron

The working can be represented as:

Inputs $\rightarrow$ Weights $\rightarrow$ Weighted Sum $\rightarrow$ Bias Addition $\rightarrow$ Activation Function $\rightarrow$ Output



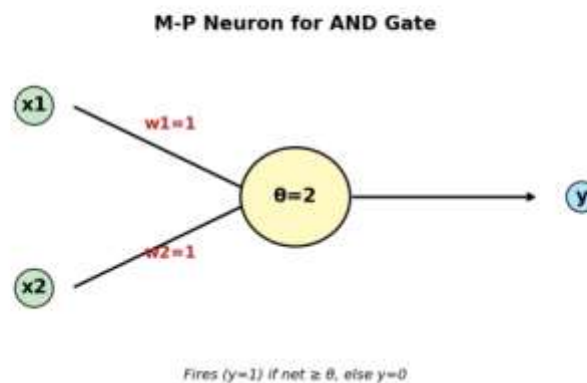
4. Construct the McCulloch–Pitts neuron model to implement the AND gate.

The McCulloch-Pitts (M-P) neuron is the simplest artificial neuron model. It takes binary inputs (0 or 1), multiplies them with fixed weights, sums them, and compares the sum with a threshold θ . If the sum $\geq \theta$, the neuron fires (output = 1); otherwise, output = 0.

Truth Table for AND Gate

x1	x2	net = w1x1+w2x2	y ($\theta=2$)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	2	1

Neuron Diagram



Design Parameters:

- **Weights:** $w_1 = 1$, $w_2 = 1$
- **Threshold:** $\theta = 2$
- **Firing Rule:** $y = 1$ if $\text{net} \geq \theta$, else $y = 0$

Formula

$$\text{net} = w_1x_1 + w_2x_2 = (1)x_1 + (1)x_2$$

Calculation for All 4 Input Combinations

- $x_1=0, x_2=0 \rightarrow \text{net} = 1(0)+1(0) = 0$; $0 \geq 2$? No $\rightarrow y = 0$
- $x_1=0, x_2=1 \rightarrow \text{net} = 1(0)+1(1) = 1$; $1 \geq 2$? No $\rightarrow y = 0$
- $x_1=1, x_2=0 \rightarrow \text{net} = 1(1)+1(0) = 1$; $1 \geq 2$? No $\rightarrow y = 0$
- $x_1=1, x_2=1 \rightarrow \text{net} = 1(1)+1(1) = 2$; $2 \geq 2$? Yes $\rightarrow y = 1$

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6. Differentiate between weights and bias in an artificial neuron with suitable examples.

Difference between Weights and Bias	
Weights	Bias
1. Weight determines the strength of connection between an input and a neuron.	1. Bias is a constant value added to the weighted sum.
2. It determines how much each input influences the output .	2. It determines when the neuron activates , independently of the input.
3. Each input generally has its own weight.	3. A neuron generally has one bias value.
4. A higher weight gives greater importance to that input.	4. Bias shifts the result towards the positive or negative side.
5. Changing the weight changes the slope of the line.	5. Changing the bias changes the position of the line.
6. Weights are adjusted during the training process.	6. Bias is also adjusted during training to improve the model.
7. A weight of zero indicates that the particular feature is insignificant.	7. Bias allows the neuron to produce a suitable output even when input values are zero.

Suitable Example Consider an artificial neuron with two inputs: $y = f(x_1w_1 + x_2w_2 + b)$ Suppose: $x_1 = 2, \quad x_2 = 3, \quad w_1 = 0.5, \quad w_2 = 0.2, \quad b = 1$ Then, $y = f[(2 \times 0.5) + (3 \times 0.2) + 1]$ $y = f[1 + 0.6 + 1] = f(2.6)$ Here: • $w_1 = 0.5$ and $w_2 = 0.2$ determine how strongly x_1 and x_2 influence the result. • $b = 1$ is the constant added to the weighted sum and shifts the final result.
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